

RRIM — Reality Representation Integrity Module

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Independent Methodological Authority

OriginID: OOF-OID-AI-RRIM-2026-06-03-0001

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Cognitive Governance Intelligence Architecture (CLIA®)

Operational Layer: Cognitive Reality Modeling Governance Layer

Governed Space: Reality Representation Integrity

Category: AI & Interpretation

Subcategory: Reality Representation Governance

Type: Cognitive Reality Modeling Module

Parent Standard: Cognitive Reality Modeling Standard (CRMS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 3 June 2026

Compatibility: OOF Methodology OS ·

Cognitive Reality Modeling Standard (CRMS) ·

Cognitive Interpretation Integrity Standard (CIIS) ·

Cognitive Reasoning Integrity Standard (CRIS) ·

Multi-Layer Truth Validation Framework (MTVF) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Reality Representation Integrity Module (RRIM) defines the structural conditions under which internal representations of environments, systems, objects, relationships, conditions, events, and operational reality remain identifiable, traceable, coherent, reconstructable, and governance-valid throughout cognition and reality-modeling processes.

RRIM governs reality representations.

The module ensures that cognition remains capable of maintaining structured representations of reality without introducing unreconstructable distortions, representation collapse, hidden dependencies, or governance-invalid reality abstractions.

A system satisfies RRIM only if:

- reality representations remain identifiable
- representation structures remain traceable
- representation dependencies remain visible
- representation coherence remains assessable
- representation degradation remains detectable

- reality representations remain operationally valid

A cognition environment that cannot identify or reconstruct its own reality representations does not satisfy RRIM.

Module Operational Role

RRIM defines the reality-representation governance layer of CRMS.

Its role is to preserve the integrity of internal representations used by cognition.

Module Operational Space

RRIM governs:

- reality representations
- world representations
- environment representations
- object representations
- relationship representations
- operational-reality representations
- representation traceability
- representation validity

The module applies wherever cognition maintains internal models of reality.

Module Function

The module applies wherever systems must preserve:

- structured reality representations
- representation traceability
- reality-model visibility
- governance-valid abstraction
- reconstructable cognition structures
- operationally reliable reality understanding

Its function is to ensure that cognition remains capable of understanding how reality is represented internally.

Minimum Implementation Framework

1. Define the Reality Representation Object

The organization must define which reality representations require governance.

This may include:

- environment models
 - object models
 - relationship models
 - operational-state models
 - strategic models
 - simulation entities
 - agent world models
 - digital-reality representations
-

2. Define Reality Representation Conditions

The system must define the conditions under which reality representations remain valid.

This includes:

- traceability requirements
- coherence requirements
- reconstruction requirements
- visibility requirements
- dependency requirements
- governance-valid representation conditions

3. Define Representation Degradation Detection Logic

The system must define how representation degradation is identified.

This may include:

- representation corruption
- hidden dependencies
- structural inconsistency
- representation drift
- abstraction failures
- model-fragmentation conditions

4. Define Operational Response or Governance Logic

The system must define governance logic for representation-integrity failures.

Governance response may include:

- representation review
- model reconstruction
- dependency analysis
- governance intervention
- escalation
- model restriction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- representation structures
- representation changes
- dependency relationships
- governance reviews
- intervention actions
- resulting cognition outcomes

A cognition environment must not remain representation-valid if materially significant reality representations cannot be identified, reconstructed, or governed.

Use Case 1 — Autonomous World Model Agent

Scenario

An autonomous agent continuously maintains an internal representation of its environment in order to navigate, reason, and make decisions.

Application

RRIM governs representation visibility, traceability, and structural integrity throughout the world-model lifecycle.

Result

The environment gains stronger model transparency, reduced representation corruption risk, and improved cognition reliability.

Use Case 2 — Digital Twin Environment

Scenario

A digital twin platform continuously maintains internal representations of physical infrastructure, operational assets, and system relationships.

Application

RRIM governs reality-representation integrity, dependency visibility, and representation reconstruction capabilities.

Result

The organization gains stronger digital-model reliability, improved operational visibility, and reduced model-fragmentation risk.

Canonical Closing Statement

Reality Representation Integrity Module (RRIM) defines the structural conditions under which internal representations of environments, systems, objects, relationships, conditions, events, and operational reality remain identifiable, traceable, coherent, reconstructable, and governance-valid throughout cognition and reality-modeling processes.

Reality models cannot remain valid if reality representations lose integrity.
Reality representation therefore becomes a foundational integrity condition of governable reality modeling.

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Canonical Source

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